Def: ($G\mathcal{T}op$)

$G\mathcal{T}op$ is a categary.

· Object: G-spaces. G-space: a space X with a conti map $a: G \times X \to X$, G with discrete top.

· Morphism: G-equivariant maps: $f: X \to Y$ with $f(g \cdot x) = g \cdot f(x)$.

Exp:

Antipodal action of C_2 on S^n. This is a free action. → no fixed point

Let V be a rep of G. Denote by S^V the one-point compatification of V. If V is an n-dim rep, then S^V is an n-dim sphere.

For $g \in G$, $x \in S^V$, $g \cdot x = \begin{cases} g \cdot x, & x \in V \\ \infty, & x = \infty \end{cases}$

$S^{(-)}$ does not define a functor $\mathcal{M}od_{\mathbb{R}[G]} \to G\mathcal{T}op$. A map $V \to W$ of reps is proper iff it is injective. So we get a functor

$$S^{(-)}: \mathcal{M}od_{\mathbb{R}[G]}^{inj} \to G\mathcal{T}op$$

↳ the subcategary of injective maps of reps.

Def: (based G-space)

A based G-space is a G-space X equipped with an equiv map $* \to X$. A map of based G-spaces is just a basepoint-preserving that is equiv.

Denote by $G\mathcal{T}op_*$ the category of based G-spaces.

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Def (induced G-space)

$H \le G$. X is an H-space. Define the induced G-space of X up to G, denoted $\uparrow_H^G(X)$ or $G \times_H X$, as the quotient of $G \times X$ by $(g \cdot h, x) \sim (g, h \cdot x)$.

Exp

$H \le G$. $\uparrow_H^G(*) = G \times_H * \cong G/H$.

If X is any space with trivial H-action, $\uparrow_H^G(X) = G/H \times X$, where G acts on G/H.

Prop

$H \le G$. Induction of spaces is left adjoint to restriction.

$$\uparrow_H^G : H\mathcal{T}op \rightleftarrows G\mathcal{T}op : \downarrow_H^G$$

$$Map_G(G \times_H X, Y) \cong Map_H(X, \downarrow_H^G Y)$$

For G-equi map $F: G \times_H X \to Y$, it is decided by values on $[e, x]$, since $[g, x] = g[e, x]$. Define $f: X \to \downarrow_H^G Y$, $f(x) = F([e, x])$. f is H-equi.

For H-equi map $f: X \to \downarrow_H^G Y$, define $F: G \times_H X \to Y$ by $F([g, x]) = g \cdot f(x)$.

Def (induced based G-space)

$H \le G$. X be a based H-space. Define the induced based G-space of X up to G, denote $\uparrow_H^G(X)$, as $G_+ \wedge_H X$, which denote the quotient of $\underbrace{G_+ \wedge X}_{(G \times X)/(G \times x_0)}$ by $(g \cdot h, x) \sim (g, h \cdot x)$.

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Prop)

$H \leq G$. Induction of spaces is left adjoint to restriction.

$$\uparrow_H^G : H\,\mathcal{T}op_* \rightleftarrows G\,\mathcal{T}op_* : \downarrow_H^G$$

$$Map_G(G_+ \wedge_H X, Y) \cong Map_H(X, \downarrow_H^G Y)$$

Def (fixed points)

For $X \in G\,\mathcal{T}op$, define the fixed points of X to be

$$X^G = \{x \in X \mid g \cdot x = x \text{ for } \forall g \in G\}$$

equipped with the subspace top as a subspace of X.

Exp)

$(S^\sigma)^{C_2} \cong S^0$. $(S^{\rho_{C_2}})^{C_2} \cong S^1$.

Prop)

The fixed points funtor is right adjoint to the trivial G-action functor.

$$triv = q^*: \mathcal{T}op \rightleftarrows G\,\mathcal{T}op : (-)^G$$

$$Map_{G\mathcal{T}op}(triv\,X, Y) \cong Map_{\mathcal{T}op}(X, Y^G)$$

Def (H-fixed points functor)

For a G-space, define H-fixed points functor as the composite

$$G\,\mathcal{T}op \xrightarrow{\downarrow_H^G} H\,\mathcal{T}op \xrightarrow{(-)^H} \mathcal{T}op$$

Exp)

$G = K_4$, $(S^{p_1^*\sigma})^L \cong (S^{p_1^*\sigma})^D \cong S^0$, $(S^{p_1^*\sigma})^R \cong S^1$.

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Def (homotopy)

Let $X \underset{g}{\overset{f}{\rightrightarrows}} Y$ be equi maps between G-spaces. Letting G act on the interval trivially, a homotopy between f and g is an equi map $h: I \times X \to Y$ that restricts to f and g at 0 and 1.

This forces each map $h_t: X \to Y$ to be equi.

Def (equi homotopy group system)

$\underline{\pi}_n(X, x_0)(G/H) = \pi_n(X^H, x_0)$, $x_0 \in X^G$.

$\underline{\pi}_n(X)(G/H) = \pi_n(X^H)$

$\mathcal{O}_G$ is is the orbit category of G:

- objects: G/H, $H \leq G$.
- morphisms: G-equi maps $G/H \to G/K$

So $\underline{\pi}_n(X): \mathcal{O}_G^{op} \to$ Groups for $n \geq 1$. If $n=0$, images are on sets.

A G-map $\alpha: G/H \to G/K$ decided by $\alpha(eH) = aK$, s.t. $a^{-1}Ha \leq K$. Then it induces a map $\alpha^*: X^K \to X^H$, $x \mapsto a \cdot x$.

For $\underline{A}, \underline{B}: \mathcal{O}_G^{op} \to$ Groups to be two system, a morphism $\eta: A \to B$ is a natural transformation, s.t.

$$\begin{tikzcd} \underline{A}(G/K) \arrow[r, "\eta_K"] \arrow[d, "\alpha^*"] \arrow[dr, phantom, "\circlearrowleft"] & B(G/K) \arrow[d, "\alpha^*"] \\ \underline{A}(G/H) \arrow[r, "\eta_H"] & \underline{B}(G/H) \end{tikzcd}$$

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Def (weak G-homotopy equivalence)

A G-equi map $f: X \to Y$ is a weak G-homotopy equivalence if it induces an iso on $\underline{\pi}_n$ for all $n \geq 0$. It induces a bijection on $\underline{\pi}_0$ and iso from $\underline{\pi}_n(X, x_0) \to \underline{\pi}_n(Y, f(x_0))$, $n \geq 1$, for each choice of basepoint $x_0 \in X^G$.

We have

$$[(G/H)_+ \wedge S^n, X]_G \cong [S^n, X^H] \cong \pi_n(X^H)$$

↳ G-equi homotopy class

For a G-map $F: (G/H)_+ \wedge S^n \to X$, define $f: S^n \to X$ by $f(s) = F(H, s)$. We can check $f: S^n \to X^H$. For a map $f: S^n \to X^H$, define $F: (G/H)_+ \wedge S^n \to X$ by $F(gH, s) = g \cdot f(s)$. We can check F is well-defined and F is a G-map.

So

$$Map_G((G/H)_+ \wedge S^n, X) \cong Map(S^n, X^H)$$

Then, we get $[(G/H)_+ \wedge S^n, X]_G \cong [S^n, X^H]$.

That means that $(G/H)_+ \wedge S^n$ can detect $\pi_n(X^H)$.

Def: (G-CW complex)

A G-CW complex is a space X built as a colimit of G-space X_n, where X_0 is a discrete G-set and X_n is obtained from X_{n-1} by attaching cells of type $(G/H) \times D^n$ (H is not fixed).

Exp:

If X is a non-equi CW-complex, we may consider it as a

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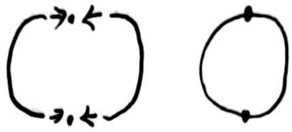
G-CW complex with trivial G-action. All cells will be of form $(G/G)\times D^n$.

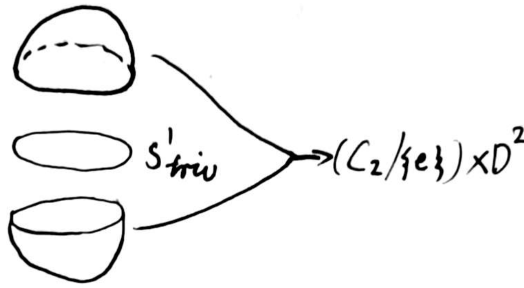
$G = C_2$. S^σ 0-cell 

1 cell $(C_2/\{e\})\times D^1$



$G = C_2$. Consider the regular rep sphere, $S^{\rho_{C_2}}$.

$X_1 = S^1_{triv}$.



Thm (equi Whitehead Theorem)

Let $f: X \to Y$ be a weak G-homotopy equivalence between G-CW complexes. Then, f is a G-homotopy equivalence.

Def (universal free G-space, classifying space of G)

G is a top group. A space EG is called a universal free G-space if

- EG is contractible,
- G acts freely on EG.

The classifying space of G, denoted by BG, is defined to be the orbit space

$$BG := EG/G.$$

Equivalently, the is a principal G-bundle

$$G \to EG \to BG$$

called the universal principal G-bundle.

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Exp

$G = C_2$. $EC_2 = S^\infty$ with antipodal action of C_2. $BC_2 \simeq S^\infty / C_2 \cong \mathbb{R}P^\infty$.

$EG \times EH = E(G\times H)$. $B(G\times H) \simeq BG \times BH$, for any $G, H \in$ Groups.

Prop

The space BG is a $K(G,1)$ meaning that $\pi_1(BG) \cong G$ and all other homotopy groups vanish.

Prop

The cellular chains $C_*(EG)$ on EG form a free resolution of $\mathbb{Z}$ over $\mathbb{Z}[G]$.

Cor

For any abelian group A, thought of as a trivial G-module, we have isos

$\hookrightarrow \mathbb{Z}[G]$-module

$$H_*(BG; A) \cong H(G; A) \text{ and } H^*(BG; A) \cong H^*(G)$$

Def: (Borel construction)

Given a G-space X, define the Borel construction on X be the orbit space of the G-action on $EG \times X$. The Borel construction is often denoted $EG \times_G X$ and we define the Borel-equi homology and cohomology of X to be

$$H_*^{Borel}(X) = H_*(EG \times_G X),\ H^*_{Borel}(X) = H^*(EG \times_G X)$$

$EG \times_G X := (EG \times X)/G$.

The G-action on $EG \times X$ is $g\cdot(e,x) = (ge, gx)$, $g \in G$, $(e,x) \in EG$.

$X_{hG} := EG \times_G X$ $\hookrightarrow$ It is also called the homotopy orbit space.

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$X^{hG} := \mathrm{Map}_G(EG, X)$, the homotopy fixed point space

Exp

a C_2-equi map : $S^1_a \to S^1$ ↳ trivial action

↓ antipodal action

If $\cdot 2$ is an iso in a abelian group A,

$$H^*_{Borel}(S^1; A) \xrightarrow{\sim} H^*_{Borel}(S^1_a; A)$$

is an iso. But S^1 and S^1_a are not C_2-equi homotopy equivalent, since their fixed points spaces are not equivalent.

So Borel homology may not tell the difference between fixed point spaces.

Def (underlying equivalence)

A G-equi map $f: X \to Y$ is an underlying equivalence if it induces an equivalence on underlying spaces.

Prop

The functor $EG \times (-) : G\mathcal{T}op \to G\mathcal{T}op$ takes underlying equivalences to G-equivalence. The homotopy orbit and homotopy fixed points constructions therefore take underlying equivalences to equivalences of spaces, and Borel cohomology takes underlying equivalences to isos.

Thm

For any G-space X and abelian group A of coefficients, the groups

$H^*_{Borel}(X)$, as H ranges over the subgroups of G, form a G-Macky functor.

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Denote $\underline{H}^*_{Borel}(X)$ as the Mackey functor.

Def (cohomological Mackey functor)

$\underline{M} \in Mack_G$ is a cohomology Mackey functor, if for each $H \leq K$, the composition

$$\underline{M}(K) \xrightarrow{R^K_H} \underline{M}(H) \xrightarrow{I^K_H} \underline{M}(K)$$

is multiplication by the index $|K:H|$.

Def (orbit category)

Let O_G denote the orbit category of G, whose objects are the G orbits G/H and whose morphisms are just the G-equiv maps.

O_G is a full subcategory of the category $FinSet_G$ of finite G-sets. The set $Hom_{O_G}(G/H, G/K)$ is nonempty iff H is contained in some conjugate of K.

Def (coefficient system)

A coefficient system for G is a functor $M: O_G^{op} \to Ablgp$.

Exp

For $n \geq 2$, G-space X, $\underline{\pi}_n(X)(G/H) := \pi_n(X^H)$ defines a coefficient system.

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Exp

G-space X, abelian group A, $n \geq 0$. $\underline{H}_n(X;A)(G/K) = H_n(X^K;A)$ defines a coefficient system.

Def

A map of G-coefficient system $C \to D$ is a natural transformation of functors. Write $\mathrm{Hom}_{\mathrm{Coeff}}(C,D)$ for the set of maps of coefficient systems.

Exp

$\underline{C}_*(X)(G/K) = C_*^{\mathrm{cell}}(X^K)$ is a chain complex. The differentials in these complexes commute with the restriction and conjugation homos.

$$\begin{tikzcd}
C_n^{\mathrm{cell}}(X^K) \arrow[r, "d_n"] \arrow[d, "R_H^K"] & C_{n-1}^{\mathrm{cell}}(X^K) \arrow[d, "R_H^K"] \\
C_n^{\mathrm{cell}}(X^H) \arrow[r, "d_n"] & C_{n-1}^{\mathrm{cell}}(X^H)
\end{tikzcd}$$

So $d_n : \underline{C}_n(X) \to \underline{C}_{n-1}(X)$ makes $\underline{C}_*(X)$ a chain complex of coefficient systems.

Def (Bredon cohomology)

X is a G-space, and M is a G-coefficient system. $C^*_{\mathrm{Coeff}}(X;M) := \mathrm{Hom}_{\mathrm{Coeff}}(C_*(X), M)$ is a cochain of abelian groups. We can define the Bredon cohomology of X with coefficients M to be

$$H^n_{Bredon}(X;M) = H^n(C^*_{Coeff}(X;M)).$$

Usually write $H^*_G(X;M)$ for Bredon cohomology.

Prop

C is a G-coefficient system. Then, $Hom_{Coeff}(C, \underline{\mathbb{Z}}) \cong Hom_{AbGp}(C(e)/G, \mathbb{Z})$.

Exp

$G = C_2$. $X = S^\sigma$. $\underline{C}_*(S^\sigma)$ is

$$\begin{tikzcd} 0 \arrow[r] \arrow[d] & \mathbb{Z}^2 \arrow[d, "id"] \\ \mathbb{Z}[C_2] \arrow[r, "\begin{bmatrix} 1 \\ -1 \end{bmatrix}"] & \mathbb{Z}^2 \end{tikzcd}$$

According to the Prop.,

$$\mathbb{Z} \xleftarrow{[1\ -1]} \mathbb{Z}^2$$

So

$$H^k_{C_2}(S^\sigma; \underline{\mathbb{Z}}) \cong \begin{cases} \mathbb{Z} & k=0 \\ 0 & k>0 \end{cases}$$

Prop

For any G-space X, $H^n_G(X;\underline{\mathbb{Z}}) \cong H^n(X/G;\mathbb{Z})$

Prop

C is a G-coefficient system. $Hom_{Coeff}(C, Inf^G_{G/G}(\mathbb{Z})) \cong Hom_{AbGp}(C(G), \mathbb{Z})$.

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Prop

For any G-space X, $H^n_G(X; \mathrm{Inf}^G_{G/G}(\mathbb{Z})) \cong H^n(X^G; \mathbb{Z})$.

$\mathrm{Hom}_{\mathrm{Coeff}}(C, \uparrow^G_e(\mathbb{Z})) \cong \mathrm{Hom}_{\mathrm{AbGp}}(C(e), \mathbb{Z})$

$H^n_G(X; \uparrow^G_e(\mathbb{Z})) \cong H^n(X; \mathbb{Z})$.

$H^n_{C_p}(X; \underline{F}) \cong H^n(X/C_p, X^{C_p}; \mathbb{Z}) \cong \tilde{H}^n((X/X^{C_p})/C_p; \mathbb{Z})$

Def

C is a G-coefficient system and N is a covariant $\mathcal{O}_G \xrightarrow{N} \mathrm{AbGp}$.
Define $C \otimes_{\mathcal{O}_G} N \in \mathrm{AbGp}$ as

$$C \otimes_{\mathcal{O}_G} N = \Big(\bigoplus_{H \le G} C(H) \otimes N(H)\Big)/\sim$$

where

$r(x) \otimes y \sim x \otimes t(y)$ for $x \in C(K)$, $y \in N(H)$ and $H \le K$

$c_g(x) \otimes y \sim x \otimes c_g(y)$ for $x \in C(K^g)$ and $y \in N(K)$.

Here r is induced by r^K_H and t is induced by t^K_H.

It can be consider the coequalizer

$$\bigoplus_{H, K \le G} \bigoplus_{\mathcal{O}(G/H, G/K)} C(K) \otimes N(H) \rightrightarrows \bigoplus_{H \le G} C(H) \otimes N(H).$$

Def: (Bredon homology)

X is a G-space. N is an $\mathcal{O}_G$-module. Define the Bredon homology of X with coefficients in N to be

$$H_n^{\mathrm{Bredon}}(X; N) = H_n(\underline{C}_*(X) \otimes_{\mathcal{O}_G} N).$$

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